

Introduction to Data Science

In a world of data space where organizations deal with petabytes and exabytes of data, the era of Big Data emerged, the essence of its storage also grew.

It was a great challenge and concern for industries for the storage of data until 2010. Now when frameworks like Hadoop and others solved the problem of storage, the focus shifted to processing of data.

Data Science plays a big role here. All those fancy Sci-fi movies you love to watch around can turn into reality by Data Science.

Nowadays its growth has been increased in multiple ways and thus one should be ready for our future by learning what it is and how can we add value to it. Without any hunches, let's dive into the world of Data Science.

What is Data Science?

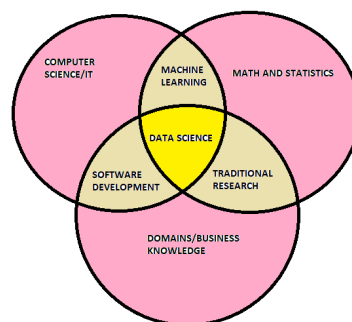
Data Science is an interdisciplinary field that focuses on extracting knowledge from data sets which are typically huge in amount. The field encompasses analysis, preparing data for analysis, and presenting findings to inform high-level decisions in an organization. As such, it incorporates skills from computer science, mathematics, statistics, information visualization, graphic, and business.

Data Science is kind of blended with various tools, algorithms, and machine learning principles. Most simply, it involves obtaining meaningful information or insights from structured or unstructured data through a process of analysing, programming and business skills.

It is a field containing many elements like mathematics, statistics, computer science, etc. Those who are good at these respective fields with enough knowledge of the domain in which you are willing to work can call themselves as Data Scientist. It's not an easy thing to do but not impossible too.

You need to start from data, its visualization, programming, formulation, development, and deployment of your model.

In the future, there will be great hype for data scientist jobs. Taking in that mind, be ready to prepare yourself to fit in this world.



How Data Science Works? Process

Data science is not a one-step process such that you will get to learn it in a short time and call ourselves a Data Scientist. It's passes from many stages and every element is important. One should

always follow the proper steps to reach the ladder. Every step has its value and it counts in your model. Buckle up in your seats and get ready to learn about those steps.

- **Problem Statement:** No work starts without motivation; Data science is no exception though. It's really important to declare or formulate your problem statement very clearly and precisely. Your whole model and it's working depend on your statement. Many scientists consider this as the main and much important step of Date Science. So, make sure what's your problem statement and how well can it add value to business or any other organization.
- **Data Collection:** After defining the problem statement, the next obvious step is to go in search of data that you might require for your model. You must do good research, find all that you need. Data can be in any form i.e. unstructured or structured. It might be in various forms like videos, spreadsheets, coded forms, etc. You must collect all these kinds of sources.
- **Data Cleaning:** As you have formulated your motive and also you did collect your data, the next step to do is cleaning. Yes, it is! Data cleaning is the most favourite thing for data scientists to do. Data cleaning is all about the removal of missing, redundant, unnecessary and duplicate data from your collection. There are various tools to do so with the help of programming in either R or [Python](#). It's totally on you to choose one of them. Various scientist has their opinion on which to choose. When it comes to the statistical part, R is preferred over Python, as it has the privilege of more than 12,000 packages. While python is used as it is fast, easily accessible and we can perform the same things as we can in R with the help of various packages.
- **Data Analysis and Exploration:** It's one of the prime things in data science to do and time to get inner Holmes out. It's about analysing the structure of data, finding hidden patterns in them, studying behaviours, visualizing the effects of one variable over others and then concluding. We can explore the data with the help of various graphs formed with the help of libraries using any programming language. In R, G G plot is one of the most famous models while Matplotlib in Python.
- **Data Modelling:** Once you are done with your study that you have formed from data visualization, you must start building a hypothesis model such that it may yield you a good prediction in future. Here, you must choose a good algorithm that best fit to your model. There different kinds of algorithms from regression to classification, SVM(Support vector machines), Clustering, etc. Your model can be of a [Machine Learning](#) algorithm. You train your model with the train data and then test it with test data. There are various methods to do so. One of them is the K-fold method where you split your whole data into two parts, one is Train and the other is test data. On these bases, you train your model.
- **Optimization and Deployment:** You followed each and every step and hence build a model that you feel is the best fit. But how can you decide how well your model is performing? This where optimization comes. You test your data and find how well it is performing by checking its accuracy. In short, you check the efficiency of the data model and thus try to optimize it for better accurate prediction. Deployment deals with the launch of your model and let the people outside there to benefit from that. You can also obtain feedback from organizations and people to know their need and then to work more on your model.

Overview of Data Science

Data science is the study of data. Like biological sciences is a study of biology, physical sciences, it's the study of physical reactions. Data is real, data has real properties, and we need to study them if we're going to work on them. Data Science involves data and some signs.

Notes of Data Science
6th Sem [AI]

Prepared By
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It is a process, not an event. It is the process of using data to understand too many different things, to understand the world. Let Suppose when you have a model or proposed explanation of a problem, and you try to validate that proposed explanation or model with your data.

It is the skill of unfolding the insights and trends that are hiding (or abstract) behind data. It's when you translate data into a story. So, use storytelling to generate insight. And with these insights, you can make strategic choices for a company or an institution.

We can also define data science as a field that is about processes and systems to extract data of various forms and from various resources whether the data is unstructured or structured.

The definition and the name came up in the 1980s and 1990s when some professors, IT Professionals, scientists were looking into the statistics curriculum, and they thought it would be better to call it data science and then later on data analytics derived.

Applications of Data Science:

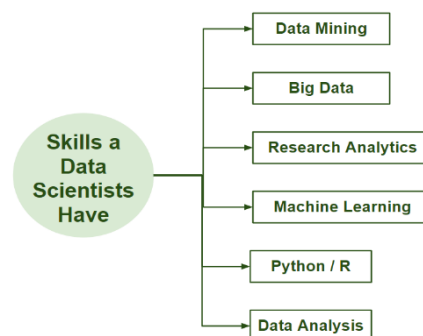
Following is some of the applications that makes use of Data Science for its services:

- Internet Search Results (Google)
- Recommendation Engine (Spotify)
- Intelligent Digital Assistants (Google Assistant)
- Autonomous Driving Vehicle (Waymo)
- Spam Filter (Gmail)
- Abusive Content and Hate Speech Filter (Facebook)
- Robotics (Boston Dynamics)
- Automatic Piracy Detection (YouTube)

Who is Data Scientist?

Is he/she someone struggling with data all day and night or experimenting in his/her laboratory with complex mathematics? After all, 'Who is a Data Scientist'?

There are many definitions available in the market for Data Scientists. In simple words, a Data Scientist is one who knows and practices the art of Data Science. The super-popular term 'Data Scientist' was coined by DJ Patil and Jeff Hammerbacher. Data Scientists are those scientists who crack complex data problems with their strong expertise in certain scientific disciplines. They work with many elements related to mathematics, statistics, probability, Quantitative and Qualitative forecasting, computer science, etc. (though they may not be an expert in all these fields).



We can say that Data Scientists are Business Analysts and Data Analysts, with a difference! Though the initial training or basic requirements are similar for all these disciplines, Data Scientists require:

- Strong Business Acumen
- Strong Communication Skills
- Exploring Big Data

Just like an agricultural scientist wants to know the percentage increase in the yield of wheat this year as compared to last year's (also the reasons associated with it) or if a financial company wants to classify its customers based on their creditworthiness (before granting loans) or whether a retail organization wants to reward extra points to its loyal customers, all need data scientists to process a large volume of both structured and unstructured data in order to make crucial business decisions.

In today's dynamic and vast world, the main challenge that today's Data Scientists face is to find solutions to the existing business problems and above it, to identify the problems that are most relevant and crucial to the organization and its success.

Key Pillars of Data Science

Usually, data scientists come from various educational and work experience backgrounds, most should be proficient in, or in an ideal case be masters in four key areas.

Pillar of data science

Domain Knowledge:

Most people thinking that domain knowledge is not important in data science but it is essential. The foremost objective of data science is to extract useful insights from that data so that it can be profitable to the company's business. If you are not aware of the business side of the company that how the business model of the company works and how you can't build it better than you are of no use for this company.

You need to know how to ask the right questions from the right people so that you can perceive the appropriate information you need to obtain the information you need. There are some visualization tools used on the business end like Tableau that help you display your valuable results or insights in a proper non-technical format such as graphs or pie charts that business people can understand.

Math Skills:

Linear Algebra, Multivariable Calculus & Optimization Technique: These three things are very important as they help us in understanding various machine learning algorithms that play an important role in Data Science.

Statistics & Probability: Understanding of Statistics is very significant as this is a part of Data analysis. Probability is also significant to statistics and it is considered a prerequisite for mastering machine learning.

Computer Science:

Programming Knowledge: One needs to have a good grasp of programming concepts such as Data structures and Algorithms. The programming languages used are Python, R, Java, Scala. C++ is also useful in some places where performance is very important.

Relational Databases: One needs to know databases such as SQL or Oracle so that he/she can retrieve the necessary data from them whenever required.

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Non-Relational Databases: There are many types of non-relational databases but mostly used types are Cassandra, HBase, MongoDB, CouchDB, Redis, Dynamo.

Machine Learning: It is one of the most vital parts of data science and the hottest subject of research among researchers so each year new advancements are made in this. One at least needs to understand basic algorithms of Supervised and Unsupervised Learning. There are multiple libraries available in Python and R for implementing these algorithms.

Distributed Computing: It is also one of the most important skills to handle a large amount of data because one can't process this much data on a single system. The tools that mostly used are Apache Hadoop and Spark. The two major parts of these tools are HDFS (Hadoop Distributed File System) that is used for collecting data over a distributed file system. Another part is map-reduce, by which we manipulate the data. One can write map-reduce in programs in Java or Python. There are various other tools such as PIG, HIVE, etc.

Communication Skill:

It includes both written and verbal communication. What happens in a data science project is after drawing conclusions from the analysis, the project has to be communicated to others. Sometimes this may be a report you send to your boss or team at work. Other times it may be a blog post. Often it may be a presentation to a group of colleagues. Regardless, a data science project always involves some form of communication of the projects' findings. So, it's necessary to have communication skills for becoming a data scientist.

Difference between Data Scientist, Data Analyst, and Data Engineer:

Data Scientist	Data Analyst	Data Engineer
The focus will be on the futuristic display of data.	The main focus of a data analyst is on optimization of scenarios, for example how an employee can enhance the company's product growth.	Data Engineers focus on optimization techniques and the construction of data in a conventional manner. The purpose of a data engineer is continuously advancing data consumption.
Data scientists present both supervised and unsupervised learning of data, say regression and classification of data, Neural networks, etc.	Data formation and cleaning of raw data, interpreting and visualization of data to perform the analysis and to perform the technical summary of data.	Frequently data engineers operate at the back end. Optimized machine learning algorithms were used for keeping data and making data to be prepared most accurately.

Data Scientist

Skills required for Data Scientist are Python, R, SQL, Pig, SAS, Apache Hadoop, Java, Perl, Spark.

Data Analyst

Skills required for Data Analyst are Python, R, SQL, SAS.

Data Engineer

Skills required for Data Engineer are MapReduce, Hive, Pig Hadoop, techniques.

Importance of data science:

- Data is a precious asset of any organization. It helps firms understand and enhance their processes, thereby saving time and money. Wastage of time and money, such as a terrible advertising decision, can deplete resources and severely impact a business.
- The efficient use of data enables businesses to reduce such wastage by analysing different marketing channels' performance and focusing on those offering the highest ROI. Thus, a company can generate more leads without increasing its advertising spend.
- The importance of Data Science lies in its innumerable uses that range from daily activities like asking Siri or Alexa for recommendations to more complex applications like operating a self-driving car.
- According to IDC, by 2025, global data will grow to 175 zettabytes. Data Science enables companies to efficiently understand gigantic data from multiple sources and derive valuable insights to make smarter data-driven decisions.
- Data Science is widely used in various industry domains, including marketing, healthcare, finance, banking, policy work, and more. That explains why Data Science is important.
- Importance of Data Science in Business

There are various reasons why Data Science is important in business. Data Science enables enterprises to measure, track, and record performance metrics for facilitating enterprise-wide enhanced decision making. Companies can analyse trends to make critical decisions to engage customers better, enhance company performance, and increase profitability.

- Data Science models use existing data and can simulate several actions. Thus, companies can devise the path to reap the best business outcomes. Data Science helps organizations identify and refine target audiences by combining existing data with other data points for developing useful insights.
- Data Science also helps recruiters by combining data points to identify candidates that best fit their company needs.
- Importance Of Data Science in IT Industry

In the present world, the generation and application of information is a critical economic activity. Data Science facilitates it with its power to extract information from large volumes of data.

- Information Technology makes our life easier by gathering and processing more data quickly and efficiently to provide results in hours contrary to days and weeks.

Big data and data Science

Big Data: It is huge, large, or voluminous data, information, or the relevant statistics acquired by large organizations and ventures. Many software and data storages is created and prepared as it is difficult to compute the big data manually. It is used to discover patterns and trends and make decisions related to human behaviour and interaction technology.

Types Of Big Data

Following are the types of Big Data:

1. **Structured**
2. **Unstructured**
3. **Semi-structured**

Structured

Any data that can be stored, accessed and processed in the form of fixed format is termed as a 'structured' data. Over the period of time, talent in computer science has achieved greater success in developing techniques for working with such kind of data (where the format is well known in advance) and also deriving value out of it. However, nowadays, we are foreseeing issues when a size of such data grows to a huge extent, typical sizes are being in the rage of multiple zettabytes.

Do you know? 10²¹ bytes equal to 1 zettabyte or one billion terabytes forms a zettabyte. Data stored in a relational database management system is one example of a 'structured' data.

Examples Of Structured Data

An 'Employee' table in a database is an example of Structured Data

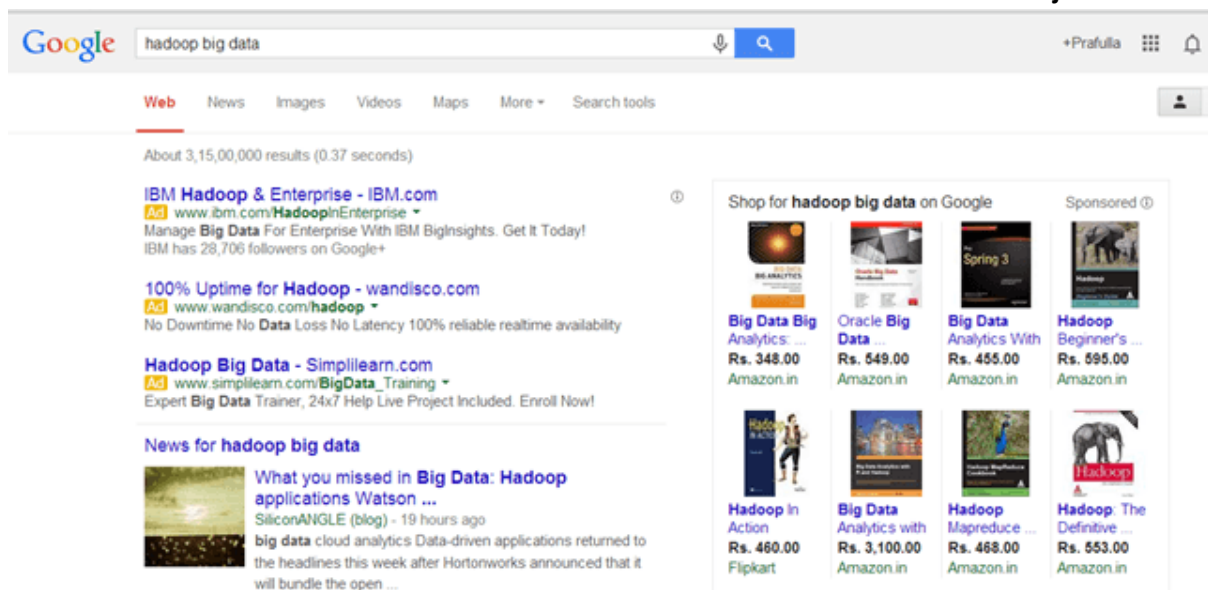
Employee_ID	Employee_Name	Gender	Department	Salary_In_lacs
5	Rajesh Kulkarni	Male	Finance	650000
8	Pratibha Joshi	Female	Admin	650000
5	Shushil Roy	Male	Admin	500000
0	Shubhojit Das	Male	Finance	500000
9	Priya Sane	Female	Finance	550000

Unstructured

Any data with unknown form or the structure is classified as unstructured data. In addition to the size being huge, un-structured data poses multiple challenges in terms of its processing for deriving value out of it. A typical example of unstructured data is a heterogeneous data source containing a combination of simple text files, images, videos etc. Now day organizations have wealth of data available with them but unfortunately, they don't know how to derive value out of it since this data is in its raw form or unstructured format.

Examples Of Un-structured Data

The output returned by 'Google Search'



Example Of Un-Structured Data

Semi-structured

Semi-structured data can contain both the forms of data. We can see semi-structured data as a structured in form but it is actually not defined with e.g., a table definition in relational DBMS. Example of semi-structured data is a data represented in an XML file.

Examples Of Semi-Structured Data

Personal data stored in an XML file-

```
<rec><name>Prashant Rao</name><sex>Male</sex><age>35</age></rec>
<rec><name>Seema R.</name><sex>Female</sex><age>41</age></rec>
<rec><name>Satish Mane</name><sex>Male</sex><age>29</age></rec>
<rec><name>Subrato Roy</name><sex>Male</sex><age>26</age></rec>
<rec><name>Jeremiah J.</name><sex>Male</sex><age>35</age></rec>
```

Characteristics Of Big Data

Big data can be described by the following characteristics:

- Volume
- Variety
- Velocity
- Variability

(i) **Volume** – The name Big Data itself is related to a size which is enormous. Size of data plays a very crucial role in determining value out of data. Also, whether a particular data can actually be considered as a Big Data or not, is dependent upon the volume of data. Hence, '**Volume**' is one characteristic which needs to be considered while dealing with Big Data solutions.

(ii) **Variety** – The next aspect of Big Data is its **variety**.

Variety refers to heterogeneous sources and the nature of data, both structured and unstructured. During earlier days, spreadsheets and databases were the only sources of data considered by most of the applications. Nowadays, data in the form of emails, photos, videos, monitoring devices, PDFs, audio, etc. are also being considered in the analysis applications. This variety of unstructured data poses certain issues for storage, mining and analysing data.

(iii) Velocity – The term ‘**velocity**’ refers to the speed of generation of data. How fast the data is generated and processed to meet the demands, determines real potential in the data.

Big Data Velocity deals with the speed at which data flows in from sources like business processes, application logs, networks, and social media sites, sensors, [Mobile](#) devices, etc. The flow of data is massive and continuous.

(iv) Variability – This refers to the inconsistency which can be shown by the data at times, thus hampering the process of being able to handle and manage the data effectively.

Advantages Of Big Data Processing

Ability to process Big Data in DBMS brings in multiple benefits, such as-

- Businesses can utilize outside intelligence while taking decisions

Access to social data from search engines and sites like Facebook, twitter are enabling organizations to fine tune their business strategies.

- Improved customer service

Traditional customer feedback systems are getting replaced by new systems designed with Big Data technologies. In these new systems, Big Data and natural language processing technologies are being used to read and evaluate consumer responses.

- Early identification of risk to the product/services, if any
- Better operational efficiency

Big Data technologies can be used for creating a staging area or landing zone for new data before identifying what data should be moved to the [data warehouse](#). In addition, such integration of Big Data technologies and data warehouse helps an organization to offload infrequently accessed data.

FUTURE OF DATA SCIENCE

Companies now have enormous datasets at their disposal thanks to the documentation of every aspect of customer interaction. The role of data science is significant in analysing and creating machine learning models based on this data. This is because these datasets are used to generate valuable insights. So, it's reasonable to assume that as analysis and machine learning improve, so will the demand for data science.

As the field expands, more jobs should become available as more data scientists are needed for analysis. Individuals wishing to pursue a career in data science can look forward to a bright future with data science. Data Science has a huge scope across all industries.

Another important aspect in the future of data science is Artificial Intelligence. AI is likely to be the most powerful technology that data scientists will have to deal with in the future. To put it another way, the future of data science will align itself to make it better in the long run. Artificial intelligence is already helping businesses make decisions and keep things running smoothly. Applied to real-world

situations, artificial intelligence will use automated solutions to screen through enormous amounts of data to uncover patterns that aid current businesses in making better decisions.

The current Scenario:-

Data Science is Transforming Businesses Worldwide

- The size of [Big data](#) is really surprising, and it has just interwoven itself in core aspects of individual and business life. Customers are getting progressively mindful of their data privacy rights and data habits, while organizations have utilized such Intel to incredible impact.
- A predominant topic today and going ahead, big data is ready to play a compelling job in the future. Information will characterize current medicinal services, government, finance, business management, marketing, energy and manufacturing.
- That implies gifted talent will be required over these businesses to address the difficulties of data analytics and help enhance improvements in products, services and society.
- The application of analytics and the utilization of machine learning tools to derive value in information, otherwise called data science, is developing and we have quite recently started to expose what's underneath.
- The three interlaced patterns of growing amounts of data, improved machine learning algorithms and better computing assets are forming the data science field in exciting manners. [Data Science](#) is only a quantitative way to deal with an issue. Before, because of the absence of information or potential processing power, we depended on different things, similar to 'authoritarian whim', 'expert intuition' and 'general consensus'.
- Today that doesn't work by any means, and, question less, it will be even less compelling a long time from now. Data scientists, thus, are building systems that can talk, foresee, envision and give real outcomes.
- The bubble around science abilities isn't set to blast. Unexpectedly, the introduction of data-driven strategies will keep on picking up predominance.
- More individuals will see information, gain bits of knowledge from it, thus it might lead to the utilization of the data science team as an indispensable part of any fruitful company or, at any rate, most of them. It might even increase the rage of competitiveness and the want to be on the top.
- It's important to concentrate on the researcher aspect of a data scientist, accepting that a data scientist must have the option to detail a question or theory from observed information, sanction and test upon this theory and along these lines arrive at a conclusion and offer their outcomes. Noticing that most data scientists are just expected to produce repeatable models, challenging your data scientists to improve and enhance is genuinely where achievement lies.
- The absence of pushing data scientists to enhance their profession's past essentially model deployment is a motivation behind why a lot of companies have difficulties with data science and AI. We see colossal impacts of data science across businesses; however, some are more developed than others, especially in finance. They're not really in the places you'd anticipate.
- We see enormous improvement being made and this is to a great extent is in light of the fact that these organizations have a ton of data as of now. Like finance has a long history of making data helpful, thus there is now a culture of being reasonably data-driven set up in a large number of these organizations, and they're additionally keen on stretching out those capabilities to new sorts of information.

- As that is where we've seen individuals beginning to make unstructured data valuable in the manners that only structured data has been helpful previously, like text Data science is working pretty intensely in the media also. That is things like understanding your crowd, helping them discover content they'll cherish, helping them draw in with that content, ensuring it's shared ideally across various platforms. It's one spot, however extremely truly distributed.
- The exponential growth in data we have seen since the start of our digital period will back off at any point soon. Truth be told, we have most likely just observed a hint of something larger. The coming years will realize a consistently expanding downpour of information.
- The new information will work as rocket fuel for our data science models, offering rise to better models as well as new and imaginative use cases. [Artificial Intelligence](#) algorithms and within the sub field of deep learning, have progressed quickly over the most recent years. What's more, there is an exceptional improvement in machine learning software. This is improving the quality of the algorithms and making the tools simpler to utilize, bringing down the barriers to entry for budding data scientists.
- In view of the solid reliance on machine learning tools, headway's in this field straightforwardly impact the value and capabilities of data science. The acknowledgment of these points of interest has driven the adoption of other AI applications, for example, machine learning and deep learning, the true future of data science.
- It goes beyond the limits of fundamental automation to deliver more prominent knowledge. Better and easier-to-use algorithms will emphatically impact data science by improving our present models and will empower the utilization of machine learning models for tasks that were recently saved for people.
- The organizations that can utilize and apply these algorithms in their business procedures will presumably build up a strong comparative advantage over their competitors.

differences between Big Data and Data Science:

Data Science

Big Data

Data Science is an area.

It is about the collection, processing, analysing, and utilizing of data in various operations. It is more conceptual.

It is a field of study just like Computer Science, Applied Statistics, or Applied Mathematics.

The goal is to build data-dominant products for a venture.

Tools mainly used in Data Science include SAS, R, Python, etc

It is a superset of Big Data as data science consists of Data scrapping, cleaning, visualization, statistics, and many more techniques.

It is mainly used for scientific purposes.

It broadly focuses on the science of the data.

Big Data is a technique to collect, maintain and process huge information.

It is about extracting vital and valuable information from a huge amount of data.

It is a technique for tracking and discovering trends in complex data sets.

The goal is to make data more vital and usable i.e., by extracting only important information from the huge data within existing traditional aspects.

Tools mostly used in Big Data include Hadoop, Spark, Flink, etc.

It is a sub-set of Data Science as mining activities which is in a pipeline of Data science.

It is mainly used for business purposes and customer satisfaction.

It is more involved with the processes of handling voluminous data.

Types of Data: -

- Structured vs. Unstructured Data
- Quantitative vs. Categorical Data
- Big Data vs. Little Data

on the basis of	Structured data	Unstructured data
Technology	It is based on a relational database.	It is based on character and binary data.
Flexibility	Structured data is less flexible and schema-dependent.	There is an absence of schema, so it is more flexible.
Scalability	It is hard to scale database schema.	It is more scalable.
Robustness	It is very robust.	It is less robust.
Performance	Here, we can perform a structured query that allows complex joining, so the performance is higher.	While in unstructured data, textual queries are possible, the performance is lower than semi-structured and structured data.
Nature	Structured data is quantitative, i.e., it consists of hard numbers or things that can be counted.	It is qualitative, as it cannot be processed and analysed using conventional tools.
Format	It has a predefined format.	It has a variety of formats, i.e., it comes in a variety of shapes and sizes.
Analysis	It is easy to search.	Searching for unstructured data is more difficult.

- Quantitative vs. Categorical Data

What is quantitative data?

Quantitative data refers to any information that can be quantified. If it can be counted or measured, and given a numerical value, it's quantitative data. Quantitative data can tell you “How many,” “how much,” or “how often”—for example, how many people attended last week's webinar? How much revenue did the company make in 2019? How often does a certain customer group use online banking?

To analyze and make sense of quantitative data, you'll conduct statistical analyses.

What is qualitative data?

Unlike quantitative data, qualitative data cannot be measured or counted. It's descriptive, expressed in terms of language rather than numerical values.

Researchers will often turn to qualitative data to answer “Why?” or “How?” questions. For example, if your quantitative data tells you that a certain website visitor abandoned their shopping cart three times in one week, you'd probably want to investigate why—and this might involve collecting some form of qualitative data from the user. Perhaps you want to know how a user feels about a particular product; again, qualitative data can provide such insights. In this case, you're not just looking at numbers; you're asking the user to tell you, using language, why they did something or how they feel.

Types of quantitative data

Discrete data is a type of numerical data with countable elements. This type of data includes incidences, proportions, or characteristics that are counted in non-negative integers. One example of this is the number of tickets in a support queue.

Continuous data is a numerical data type with uncountable elements. It can be any value (no matter how big or small) measured on a limitless scale. For example, the measure of time and temperature are continuous. Continuous data can be further.

- **Interval data** can be measured along a continuum, where there is an equal distance between each point on the scale. Interval data has no true or meaningful zero value. For instance, the difference between 5 and 6 feet is equal to the difference between 25 and 50 miles on a scale.
- **Ratio data** is similar to interval data in that it's equally spaced on a scale, but unlike interval data, ratio data has a [true zero](#). Weight is classified as ratio data; whether it has equal weight or weighs zero grams—it weighs nothing at all.

Types of qualitative (Categorical data)

Nominal data Nominal data is used to name variables without providing numerical value. Nominal data is sometimes referred to as “named” data. Examples of nominal data include name, height, and weight.

Ordinal data Ordinal data has a set order or scale to it. It can be both types of data, but it exhibits more categorical data characteristics. Some examples of ordinal variables include [customer satisfaction surveys](#), interval scales, and bug escalation.

Quantitative data:

- My best friend is 5 feet and 7 inches tall
- They have size 6 feet
- They weigh 63 kilograms
- My best friend has one older sibling and two younger siblings
- They have two cats

- My best friend lives twenty miles away from me
- They go swimming four times a week

Qualitative data:

- My best friend has curly brown hair
- They have green eyes
- My best friend is funny, loud, and a good listener
- They can also be quite impatient and impulsive at times
- My best friend drives a red car
- They have a very friendly face and a contagious laugh

BASIS FOR COMPARISON	QUALITATIVE DATA	QUANTITATIVE DATA
Meaning	Qualitative data is the data in which the classification of objects is based on attributes and properties.	Quantitative Data is the type of data which can be measured and expressed numerically.
Research Methodology	Exploratory	Conclusive
Approach	Subjective	Objective
Analysis	Non-Statistical	Statistical
Collection of data	Unstructured	Structured
Determines	Depth of understanding	Level of occurrence

BASIS FOR COMPARISON	QUALITATIVE DATA	QUANTITATIVE DATA
Asks	Why?	How many or How much?
Sample	Small number of non-representative samples	Large number of representative samples
Outcome	Develops initial understanding.	Recommends final course of action.

Small Data: It can be defined as small datasets that are capable of impacting decisions in the present. Anything that is currently ongoing and whose data can be accumulated in an Excel file. Small Data is also helpful in making decisions, but does not aim to impact the business to a great extent, rather for a short span of small data can be described as small datasets that are capable of having an influence on current decisions. Almost everything currently in progress and the data of which can be acquired in an Excel file. Small data is also useful in decision-making but is not intended to have a large impact on business, rather for a short period of time.

In nutshell, data that is simple enough to be used for human understanding in such a volume and structure that makes it accessible, concise, and workable is known as small data.

Big Data: It can be represented as large chunks of structured and unstructured data. The amount of data stored is immense. It is therefore important for analysts to thoroughly dig the whole thing into making it relevant and useful to make proper business decisions.

In short, datasets that are really huge and complex that conventional data processing techniques cannot manage them are known as big data.

Feature	Small Data	Big Data
Technology	Traditional	Modern
Collection	Generally, it is obtained in an organized manner than is inserted into the database	The Big Data collection is done by using pipelines having queues like AWS Kinesis or Google Pub / Sub to balance high-speed data
Volume	Data in the range of tens or hundreds of Gigabytes	Size of Data is more than Terabytes
Analysis Areas	Data marts(Analysts)	Clusters(Data Scientists), Data marts(Analysts)

Feature	Small Data	Big Data
Quality	Contains less noise as data is less collected in a controlled manner	Usually, the quality of data is not guaranteed
Processing	It requires batch-oriented processing pipelines	It has both batch and stream processing pipelines
Database	SQL	NoSQL
Velocity	A regulated and constant flow of data, data aggregation is slow	Data arrives at extremely high speeds, large volumes of data aggregation in a short time
Structure	Structured data in tabular format with fixed schema(Relational)	Numerous variety of data set including tabular data, text, audio, images, video, logs, JSON etc.(Non Relational)
Scalability	They are usually vertically scaled	They are mostly based on horizontally scaling architectures, which gives more versatility at a lower cost
Query Language	only Sequel	Python, R, Java, Sequel
Hardware	A single server is sufficient	Requires more than one server
Value	Business Intelligence, analysis and reporting	Complex data mining techniques for pattern finding, recommendation, prediction etc.
Optimization	Data can be optimized manually (human powered)	Requires machine learning techniques for data optimization
Storage	Storage within enterprises, local servers etc.	Usually requires distributed storage systems on cloud or in external file systems

Feature	Small Data	Big Data
People	Data Analysts, Database Administrators and Data Engineers	Data Scientists, Data Analysts, Database Administrators and Data Engineers
Security	Security practices for Small Data include user privileges, data encryption, hashing, etc.	Securing Big Data systems are much more complicated. Best security practices include data encryption, cluster network isolation, strong access control protocols etc.
Nomenclature	Database, Data Warehouse, Data Mart	Data Lake
Infrastructure	Predictable resource allocation, mostly vertically scalable hardware.	More agile infrastructure with horizontally scalable hardware

Data Science Process

Data Science could be a space that incorporates working with colossal sums of information, creating calculations, working with machine learning and more to come up with trade insights. It incorporates working with the gigantic sum of information. Different processes are included to infer the information from the source like extraction of data, information preparation, model planning, model building and many more. The below image depicts the various processes of Data Science.



- **Information Preparation**

In this stage, you would like to investigate, pre-process and condition data for modelling. You'll be able to perform information cleaning, changing, and visualization. This will assist you to spot the exceptions and build up a relationship between the factors. Once you have got cleaned and arranged the information, it's time to do exploratory analytics on it.

- **Model Planning**

Here, you may decide the strategies and methods to draw the connections between factors. These connections will set the base for the calculations which you may execute within the following stage. You may apply Exploratory Data Analytics (EDA) utilizing different factual equations and visualization apparatuses.

- **Model Building**

In this stage, you'll create datasets for training and testing purposes. You may analyse different learning procedures like classification, association, and clustering and at last, actualize the most excellent fit technique to construct the show.

- **Operationalize**

In this stage, you convey the last briefings, code, and specialized reports. In expansion, now a pilot venture is additionally actualized in a real-time generation environment. This will give you a clear picture of the execution and other related limitations.

- **Communicate Results**

Presently, it is critical to assess the outcome of the objective. So, within the final stage, you recognize all the key discoveries, communicate to the partners and decide in the event that the outcomes about the venture are a victory or a disappointment based on the criteria created in Stage 1.

Data Scientist Roles and Responsibilities

Data scientists collaborate closely with business leaders and other key players to comprehend company objectives and identify data-driven strategies for achieving those objectives. A data scientist's job is to gather a large amount of data, analyse it, separate out the essential information, and then utilize tools like SAS, R programming, Python, etc. to extract insights that may be used to increase the productivity and efficiency of the business. Depending on an organization's needs, data scientists have a wide range of roles and responsibilities. The following is a list of some of the data scientist roles and responsibilities:

- Collect data and identify data sources
- Analyse huge amounts of data, both structured and unstructured
- Create solutions and strategies to business problems
- Work with team members and leaders to develop data strategy
- To discover trends and patterns, combine various algorithms and modules
- Present data using various data visualization techniques and tools
- Investigate additional technologies and tools for developing innovative data strategies
- Create comprehensive analytical solutions, from data gathering to display; assist in the construction of data engineering pipelines
- Supporting the data scientists, BI developers, and analysts' team as needed for their projects
Working with the sales and pre-sales team on cost reduction, effort estimation, and cost optimization
- To boost general effectiveness and performance, stay current with the newest tools, trends, and technologies
- collaborating together with the product team and partners to provide data-driven solutions created with original concepts
- Create analytics solutions for businesses by combining various tools, applied statistics, and machine learning
- Lead discussions and assess the feasibility of AI/ML solutions for business processes and outcomes
- Architect, implement, and monitor data pipelines, as well as conduct knowledge sharing sessions with peers to ensure effective data use

How to become a data scientist?

The majority of data scientists begin their careers with a Bachelor's degree in mathematics, statistics, computer science, information technologies, data analytics or data science. Online courses may be required to learn the specific skills required to obtain a higher post-graduate degree in data science for those who choose to complete an undergraduate degree in a field other than data science.

In order to be considered for most data science and data analytics jobs or clear the data science interview questions, candidates must hold at least a Master's degree. Data scientists will continue to acquire and perfect new programming languages, database architecture, and other advanced data organizing and analytics abilities throughout their post-graduate education in order to succeed professionally.

While pursuing their post-graduate degrees, data scientists may be expected to take up internships to network and learn the ins and outs of their chosen sector. You could also prepare for some data science interview questions. Some people also decide to enrol in specialized courses in fields like business, physics, or biotechnology that are connected to the industry they wish to work in.